

Lab 1: MNIST-AE

Working group assignment

Submit your notebook (within a zip file) to
<http://deei-mooshak.ualg.pt/~jvo/GenAI/submissions/>
as TP1

up to: Oct. 6, 2026

Questions

1. What is a generative model?
2. What is the goal of an autoencoder?
3. What's the point for a complete autoencoder?
4. Is an autoencoder a supervised or an unsupervised learning method?
5. Given the autoencoder

Input: $x^{(i)}$

Encoder weights: W

Encoder bias: b

Hidden representation: h

Decoder weights: W^*

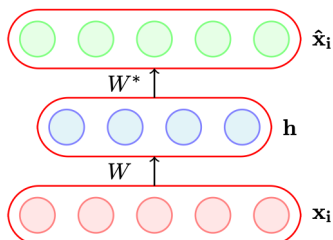
Decoder bias: c

First activation function: g

Second activation function: f

What is the formula for computing h and the output $\hat{x}^{(i)}$

6. Consider the following under complete AE



Compute the update expressions for the elements of W , for i) the vanilla AE, ii) weight decay regularized AE, and iii) sparse AE.

Empirical analysis

The companion notebook `MNIST_AE.ipynb` describes how to define and train a shallow autoencoder for (a variant of) MNIST digits using Tensorflow and Keras.

In this lab assignment, you should analyse empirically the performance of AE as a function of the number of layers. Different types of regularization should also be considered.

Presentation

Please note that your team will present your work (i.e., code, results, and conclusions) to the class in the lab session immediately after the deadline. A PowerPoint presentation should be prepared for the presentation and submitted afterwards. The presentation should not exceed 15 minutes, followed by a short period of Q&A